

Midterm Exam
ECON1002, Econ Principles of Macro

July 23rd, 2024

You have 100 minutes to complete this exam.

In the pursuit of the high ideals and rigorous standards of academic life, I commit myself to respect and uphold the Georgetown University Honor System:
To be honest in any academic endeavor, and To conduct myself honorably, as a responsible member of the Georgetown community, as we live and work together.
I agree that any code violations will result in a fail for this course.

Name: _____

Signature: _____

Question:	1	2	3	4	5	Total
Points:	6	12	20	45	17	100
Score:						

Instructions: Read all directions carefully and write your answers in the space provided. To receive full credit, you must show ***all*** of your work. Answers without supporting work will not be given credit. Make sure to sign the honor pledge statement above and to write your name. Make sure that you have all the exam pages.

1. Definition, and Short Description (6 points).

3 (a) Provide a definition of natural level of output.

3 (b) Calculate the real interest rate when the anticipated inflation rate is 2% and the nominal interest rate is 6%.

2. Answer true, false, or uncertain for the following statements **AND** justify your answer with a brief explanation. (12 points)

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(a) If planned investment I is exogenous, IS curve is horizontal on a Yr -plane.

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(b) Assume an economy with households, firms, and the government. According to the Keynesian cross model, an increase of one unit in government spending leads to an increase of one unit in equilibrium output.

3. Georgetown is a small island nation with a simple economy that produces only six goods: broccoli, tomatoes, eggs, pizzas, milk, and cheeses. Assume that one-quarter of all the broccoli and one-fifth of all tomatoes are used to produce pizzas and one-half of all the milk are used to produce cheeses. (20 points)

	2022		2023		2024	
Product	Quantity	Price	Quantity	Price	Quantity	Price
Broccoli	240	\$0.80	240	\$1.00	300	\$1.15
Tomatoes	600	2.50	700	3.00	800	4.00
Pizzas	150	10.00	160	12.00	180	15.00
Eggs	500	2.00	450	2.50	450	2.50
Milk	60	4.50	70	4.50	80	5.00
Cheeses	100	25.00	85	30.00	85	30.00

- 5 (a) Use the production and price information in the table to calculate nominal GDP for 2022, 2023, and 2024.
- 5 (b) Use the production and price information in the table to calculate real GDP for 2022, 2023, and 2024 using 2022 as the base year. What is the growth rate of real GDP from 2022 to 2023 and from 2023 to 2024?

- 5 (c) Compute the GDP deflator in 2022, 2023 and 2024 by using 2022 as the base year. Determine the inflation rate in 2023 and 2024 from the GDP deflators.
- 5 (d) Calculate the CPI in 2022, 2023, and 2024 using the production information in 2022 as the fixed market basket. Determine the inflation rate in 2023 and 2024 from the CPIs.

4. Assume the following economy of a country (45 points):

- Consumption function: $C = 100 + 0.75Y_d$
- Investment: $I = 100$
- Government spending: $G = 100$
- Net taxes: $T = 100$
- Disposable income: $Y_d \equiv Y - T$

5 (a) Derive the (planned) aggregate expenditure function AE in this economy. Draw a graph showing AE .

3 (b) Provide the equilibrium condition.

5 (c) Solve for the equilibrium income and denote the equilibrium point as E on the graph that you drew in part (a).

- 7 (d) Now suppose that government spending G has increased by 50, which means that $G' = 150$. Solve for the new equilibrium income. Represent this new equilibrium induced by fiscal policy change as a point A on a graph that you drew in part (a).
- 3 (e) Find the government spending multiplier.
- 7 (f) Now suppose that tax revenues depend on income with $G = 100$.
- Net taxes: $T = -100 + \frac{1}{3}Y$
- Solve for the equilibrium level of income and plot a graph showing the equilibrium point denoting as B .

- 7 (g) Again, suppose that government spending G has increased by 50, which means that $G' = 150$. Solve for the new equilibrium income with income tax. Represent the new equilibrium induced by fiscal policy change as a point C on a graph that you drew in part (f).
- 8 (h) Find the government spending multiplier when tax depends on income. Compare the multiplier that you obtained in part (e) and explain why they are same or different.

5. The following questions are about IS-LM & AS/AD model. Illustrate each of the following situations with graphs showing the Keynesian cross, money market, the $IS - LM$ curves, and AS/AD curves, and explain what happens to the equilibrium values of the interest rate, aggregate output, and price level (17 points).

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- (a) What do IS curve and LM curve (Fed rule) mean?

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- (b) A increase in G with the money supply held constant by the Fed

This page is intentionally left blank to accommodate work that wouldn't fit elsewhere and/or scratch work.